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State Finished

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Time taken 55 mins 5 secs

Question 1

Complete

Marked out of 1.00

Let $PA = LU$ be the PLU decomposition of matrix $A = \begin{bmatrix} 4 & 2 & 5 \\ 6 & 3 & 6 \\ 2 & 2 & 4 \end{bmatrix}$. Then $P + L =$

☐ A.

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

☒ B.

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

☐ C.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

☐ D.

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Question 2

Complete

Marked out of 1.00

In Newton-Cotes formulae for integration, we subdivide the range of integration $[a, b]$ in n equi-spaced parts.

☒ True

☐ False
Question 3

Complete

Marked out of 1.00

Consider the following data:

x	1	2	3	4	5
f(x)	2	6	12	20	30

The value of $\nabla^3 f(5)$ is given by:

☐ A. 4

☒ B. 0

☐ C. None of these

☐ D. 2
Question 4

Complete

Marked out of 1.00

Consider the following data:

x	1	2	3	4	5
f(x)	2	6	12	20	30

The value of $\Delta f(1)$ is

☒ A. 4

☐ B. None of these

☐ C. 0

☐ D. 2

Question 5

Complete

Marked out of 1.00

The inverse of a permutation matrix is its transpose.

☒ True

☐ False
Question 6

Complete

Marked out of 1.00

If $P_n(x)$ is Legendre Polynomial of degree n , the $\int_{-1}^1 P_2(x)P_3(x)dx = 0$

☒ True

☐ False
Question 7

Complete

Marked out of 1.00

Using Gauss elimination method for the following system of equation:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 13 \\ 20 \end{bmatrix} \text{ reduces the coefficient matrix to:}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

☒ True

☐ False

Question 8

Complete

Marked out of 1.00

Consider the system of algebraic equations $Ax = 0$, and $|A| \neq 0$. Then

- ☒ A. $x = 0$ is only solution
- ☐ B. There may be many solutions
- ☐ C. Information is incomplete
- ☐ D. There is no solution

Question 9

Complete

Marked out of 1.00

For an 'upper triangular matrix' $A = [a_{ij}]_{n \times n}$, the maximum number of non-zero entries is:

- ☐ A. $n^2 + n$
- ☐ B. n^2
- ☒ C. $\frac{1}{2}(n^2 + n)$
- ☐ D. None of these

Question 10

Complete

Marked out of 1.00

Consider the integral $I = \int_0^4 x^4 \cos \frac{\pi}{4} x \, dx$. The value of the integral, using Simpson's 1/3rd method, and choosing $h = 1$, is given as (approximately):

- ☐ A. None of these
- ☐ B. 160.758
- ☒ C. -482.274
- ☐ D. 160.758

Question 11

Complete

Marked out of 1.00

The minimum number of points on a plane required to fit a parabola (a quadratic polynomial) is

- ☐ A. None of these
- ☐ B. 2
- ☒ C. 3
- ☐ D. 4

Question 12

Complete

Marked out of 1.00

Choose $A = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix} = LU$, where, $L = \begin{bmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{bmatrix}$ and $U = \begin{bmatrix} 1 & u_{12} \\ 0 & 1 \end{bmatrix}$. Then, which among the following is correct:

- ☐ A. $\ell_{11} = 1, \ell_{21} = 2, \ell_{22} = 0.5, u_{12} = 5$
- ☐ B. $\ell_{11} = 6, \ell_{21} = 2, \ell_{22} = 0.5, u_{12} = 5$
- ☐ C. $\ell_{11} = 6, \ell_{21} = 2, \ell_{22} = 1, u_{12} = 5$
- ☒ D. $\ell_{11} = 6, \ell_{21} = 2, \ell_{22} = 4, u_{12} = 0.5$

Question 13

Complete

Marked out of 1.00

Using 'three-point Gauss Hermite Method', the APPROX integral of $\int_{-\infty}^{\infty} e^{-x^2} \frac{1}{x^2+1} dx$ is given by:

- ☐ A. 0.712
- ☐ B. 1.43
- ☒ C. 1.4179
- ☐ D. None of these

Question 14

Complete

Marked out of 1.00

The closest value of the integral $I = \int_{-1}^1 \frac{2x^2}{(x^2+1)^2} dx$, by using two-point Gauss Legendre rule, correct upto 3 decimal points is:

- ☐ A. None of these
- ☐ B. 0.362
- ☒ C. 0.750
- ☐ D. 0

Question 15

Complete

Marked out of 1.00

A matrix $A = [a_{ij}]_{n \times n}$ is said to be 'lower triangular matrix' if:

- ☐ A. $a_{ij} = 0, \forall i > j$
- ☐ B. None of these
- ☐ C. $a_{ij} = 0, \forall i + 1 > j$
- ☒ D. $a_{ij} = 0, \forall i < j$

Question 16

Complete

Marked out of 1.00

Let $A = LU$ be the LU decomposition of matrix $A = \begin{bmatrix} 4 & 2 & 5 \\ 1 & 1 & 2 \\ 2 & 1 & 2 \end{bmatrix}$ and U^T denotes transpose of matrix U . Then $U^T - L =$

☒ A.

$$\begin{bmatrix} 3 & 0 & 0 \\ 1.75 & -0.5 & 0 \\ 4.5 & 0.75 & -1.5 \end{bmatrix}$$

☐ B.

$$\begin{bmatrix} 3 & 0 & 0 \\ 1.75 & 0.5 & 0 \\ 4.5 & 0.75 & 1.5 \end{bmatrix}$$

☐ C.

$$\begin{bmatrix} 3 & 0 & 0 \\ -1.75 & 0.5 & 0 \\ -4.5 & -0.75 & 1.5 \end{bmatrix}$$

☐ D.

$$\begin{bmatrix} -3 & 0 & 0 \\ -1.75 & 0.5 & 0 \\ -4.5 & -0.75 & 1.5 \end{bmatrix}$$

Question 17

Complete

Marked out of 1.00

In Gauss elimination method the 'pivot' is the smallest element in the column:

☐ True

☒ False

Question 18

Complete

Marked out of 1.00

Every invertible matrix A has PLU decomposition.

☒ True☐ False

Question 19

Complete

Marked out of 1.00

The integral $I = \int_0^2 \frac{x}{(x^2+1)} dx = \int_{-1}^1 \frac{x+1}{(x^2+2x+2)} dx$:

☒ True☐ False

Question 20

Complete

Marked out of 1.00

Consider the following data:

x	1	2	3	4	5
f(x)	2	6	12	20	30

Then, $f(1.5)$ is given by

☐ A. None of these☐ B. 5☒ C. 4☐ D. 3

Question 21

Complete

Marked out of 1.00

Consider the following data:

x	1	2	3	4	5
f(x)	2	6	12	20	30

The value of $f(4.5)$ is given by:

- ☐ A. 24
- ☒ B. 25
- ☐ C. None of these
- ☐ D. 24.75

Question 22

Complete

Marked out of 1.00

Consider the following data:

x	1	2	3	4	5
f(x)	2	6	12	20	30

The polynomial of degree less than or equal to 3 that satisfy above data is given by $x^2 + x$:

☒ True☐ False**Question 23**

Complete

Marked out of 1.00

Consider the integral $I = \int_0^{1.5} e^{-x^2} dx$. The value of the integral, using Trapezoidal method, and choosing $h = 0.25$, is given as (approximately):

- ☐ A. 0.3946
- ☐ B. None of these
- ☐ C. 1.484
- ☒ D. 0.74295

Question 24

Complete

Marked out of 1.00

The closest value of the integral $I = \int_{-1}^1 \frac{2}{(x^2+1)^2} dx$, by using three-point Gauss Legendre rule, correct upto 4 decimal points is:

- ☐ A. 1.7569
- ☐ B. 0.6644
- ☐ C. 1.5769
- ☒ D. None of these

Question 25

Complete

Marked out of 1.00

To apply Simpson's 3/8th method on $I = \int_a^b f(x) dx$, we must divide the domain of the integral $[a, b]$ in n parts, where n is:

- ☐ A. even number
- ☐ B. $2m, m \in \mathbb{N}$
- ☐ C. odd number
- ☒ D. $3m, m \in \mathbb{N}$

